

Identification of bumblebee species based on their wings

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Abstract—Bumblebees play a vital role in maintaining ecological balance and agricultural productivity through their crucial role in pollination. However, the conservation of bumblebee populations presents challenges due to their cryptic nature, making species identification difficult without harming the bees. In this work, we improve the deep learning approach proposed by V. Exbrayat to predict bumblebee species instantaneously using wing images and landmarks detection. For this purpose, we have adapted Exbrayat’s technique, investigating the role of the degree for ground truth peaks shape. We have also adapted data preparation on annotated base, and designed a new data preparation algorithm to make predictions easier on unannotated base. We show that higher degree often lead to better performances on accuracy and mean error on annotated base. We developed a finetuning technique allowing to reach very high degree, leading to significant performance improvement compared to Exbrayat’s results. We reached 97 % accuracy and 0.83 pixels mean error on test set, widely outperforming precedent works. We finally managed to test our model on unannotated dataset, which gives promising results, with all prepared images apparently correctly annotated after visual examination.

Index Terms—Point prediction, UNet, Bumblebee, Wing, Morphometry

PRELIMINARY REMARKS

This work is the continuation of V. Exbrayat’s work [1], so databases, context elements, and techniques used are roughly the same. In order to make an independent article, some elements from Exbrayat’s article are reported below.

I. INTRODUCTION

(The following paragraphs come from Exbrayat’s article)

Bumblebee holds a vital role in maintaining ecological balance and supporting agricultural productivity. As species crucial for the environment, bumblebees contribute signifi-

cantly to pollination, ensuring the reproduction of various plant species and the sustainability of ecosystems. However, the complexity of bumblebee populations presents challenges for their conservation. Bumblebees are cryptic species, meaning they display diverse genetic characteristics within a single species, making identification more difficult. For some species, bumblebees need to be anesthetized before being examined by a scientific expert in a laboratory to identify from which species they belong to. Even if new anesthesia methods are being implemented [2], most of the time bees do not survive.

The inability to identify species on the ground, without killing them, raises challenges for the census and the protection of these crucial pollinators. To help scientists for the identification of these species, our idea was to use deep learning algorithms to predict instantaneously species based on bumblebee wings pictures which could be taken on the ground. Wings being unique and immutable species indicators for bees, complex biometric algorithm such as an adapted FaceNet for bumblebees could have been implemented for this work.

However, we decided to draw our inspiration from what is done currently by scientists. They annotate 18 landmarks corresponding to the veins intersections on the wings. The annotation is not randomly done at an intersection : the landmarks is annotated at the same relative position at the junction. This technique depends on specific conventions and have to be the same for all the studied wings.

Then, using a morphometric software, scientists can predict the species bumblebees belongs to. [3] Detecting the 18 landmarks on a bumblebee wing suffice to identify it. Reducing the complexity of identification algorithm, this proven method is our objective for the rest of the study given the accessibility to dataset suitable for this work.

II. RELATED WORK

(The 2 following paragraphes come from Exbrayat's article)

Some applications existed for bees identification based on wings images such as the Deep Automated Bee identification system [4]. This software offers accurate results and is downloadable on smartphone for a direct use on the ground. However, some species are more complex than others, their morphological similarities raise challenges for their identification. In the case of the lizards, tests were done to check identification for such species [5]. Results are encouraging, experts work could be lightened for the identification of cryptic species. Computer vision has a pivotal role to play in the scientific community.

Concerning bees identification, experts had to collect the 18 landmarks from wings which is tedious and time-consuming before loading the coordinates in a morphometric algorithm identifying the specie. The morphometric analysis is based on the comparison of different conformations coming from the 18 landmarks of different wings from different species. This technique is explained in Adrien PERRARD's Thesis [6].

In his work, V. Exbrayat used UNet architecture [7] and an adapted ground truth, a pseudo-density map with polynomial shaped peaks at points positions, with one channel and the same size as input image. This takes its inspiration from what is done by Lempitsky et al. [8] to count objects in images. For point recuperation, he uses h -maxima, with h value determined automatically for each prediction in order to retrieve 18 predicted annotations. Our goal is to improve Exbrayat's work, in order to get a reliable algorithm to predict annotations on bee wing pictures.

III. APPROACH

A. Datasets Overview (partly copied from Exbrayat's article)

Two datasets were at our disposal

- The first from the Institute of Mons of 1700 pictures of bumblebees wings with the 18 corresponding points (we will call this database "base 1")
- The second from the IEES (the Institute of Ecology and Environmental Sciences of Paris) of 2900 pictures of bumblebees wings belonging to different species but without annotations (we will call this database "base 2").

Therefore, with these 18 landmarks, the first dataset has been set for learning. Before using the first dataset, we made a manual sort to remove some images of bad quality, and we end up with 1717 images.

The final objective is to predict species from pictures of the second dataset which has a specific and precise way of taking pictures in field: wings open on an opaque paper, always in the same orientation. Resolution varies, but the way photos are taken not. This means that we have only few variability due to changing light or wet wings. Conversely, the pictures from the first dataset have been taken in a laboratory on a transparent paper. Some wings are wet, which leads to more light reflection. Color is variable, with apparently some photos taken under white light whereas others were taken under yellow

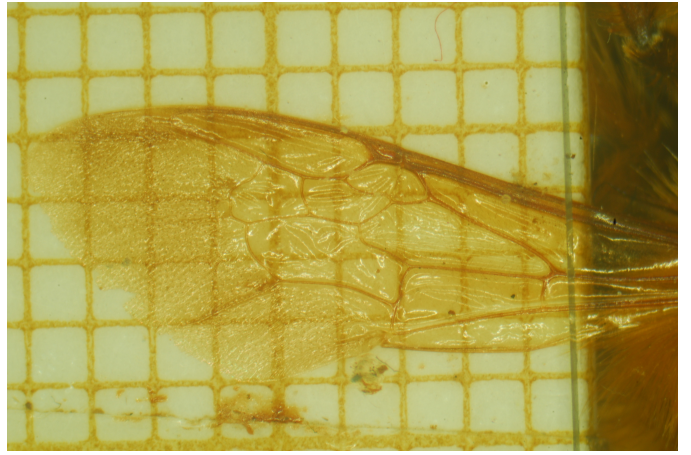


Fig. 1: Example from base 1 dataset

light. The orientation and angle of wings also vary over the first dataset. For the first dataset, most photos are taken on a grid background.



Fig. 2: Example from base 2 dataset

B. Approach Overview

Given that this work directly follows the work of V. Exbrayat, with the goal of pursuing his work, and improving performances, we decided to use the same strategy, and evaluate the roles of each parameter to find a better solution. The global approach will thus be the same as V. Exbrayat, with the same network architecture (UNet, with an input of size $256 \times 512 \times 3$, and an output of size $256 \times 512 \times 1$), the same strategy for ground truth (pseudo-density map with polynomial shaped peaks), and the same performances evaluation (evaluating the correctness of a prediction using a threshold parameter, that we will be set to 5 to fit his work, even if the choice of this parameter should be subject to discussion). The following sections will no more be adapted from Exbrayat's work, as they would go further into details.

C. Dataset preparation

We will first present the preparation of base 2 images (unannotated images), as this provides arguments to explain the preparation of the first dataset.

To normalize the wings used as input for our model, the first step is to extract the wing from the image, rotate it, and crop it. For this purpose, we used Segment Anything Model [9] from Meta, with the following parameters:

- Model used: vit_l
- Class used: SAMAutomaticMaskGenerator
- Parameters: min_mask_region_area=5000, other parameters are default (16 points_per_side for the input)

This allows us to extract all objects from an image, filtering out small noisy components. To identify the mask corresponding to the bee wing, we rotate each mask to maximize the width/height ratio, and select the largest mask with a ratio between 2.2 and 2.8 (empirically determined, could be refined).

Although this method is not fully reliable, this was enough to prepare a relevant sample from base 2, but training a CNN to detect wings would be a better long-term solution for developing a robust preprocessing tool.

After extracting the rotated wing (now nearly horizontal), we reshape the crop to fit the network architecture, targeting a width/height ratio of 2 (see Subsection III.D), by extending the bounding box along the y-axis.



Fig. 3: Preprocessed wing image from base 2

With the above process, we can prepare the wings from base 2 with close crop and almost horizontal wing.

We can thus prepare data from base 1 for training, and avoid some useless data variability. First, we rotate the images using the 2 points in the top right cell (points 0 and 1), to put all wings within a $[-7, +7]$ degrees range. For this purpose, we keep the current angle if it belong to the given angle range. Conversely, we rotate the wing to set this angle to 0° if the original angle does not belong to the given range. We then use the relative position of point 0 compared to point 1 to determine if we flip the image, so every wing got the same orientation.

Then, we crop the wings in the images using annotations, adding some randomness in the crop to avoid having some

landmarks always in the same place. We set a minimal margin of 20 % of the size of the image for each axis (eg 20 % on top, 20 % on bottom etc...), and a 30 % random additionnal margin. We finally add a additionnal margin to fit network architecture.

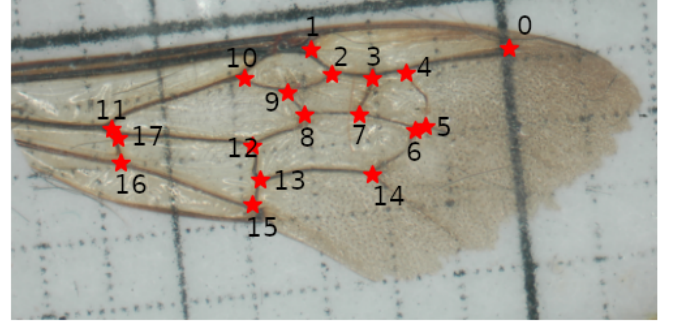


Fig. 4: Preprocessed wing image from base 1, with annotations

D. Ground Truth

As mentionned above, the ground truth that will be adopted for this work will be based on the same model as Exbrayat's work. However, our new definition bring a new glance on the subject, showing new parameters, and being more clear and adaptable that Exbrayat's one. We can define the ground truth function as follow :

for $x \in \mathbb{R}^2$,

$$F(x) = \max_{i \in [0, 17]} (f_i(\min(\frac{\|x - a_i\|_2}{r}, 1))), r \in \mathbb{R}_+^* \quad (1)$$

with $a_i, i \in [0, 17]$ the annotations, $f_i, i \in [0, 17]$ continuous decreasing functions defined on $[0, 1]$, with $f_i(0) = 1$ and $f_i(1) = 0$

This definition is of course very general and is not really useful without some adaptations. For our study, we decided to choose f_i as follow :

$$f_i(u) = (1 - u)^d, i \in [0, 17], u \in [0, 1], d \in \mathbb{N}^* \quad (2)$$

This is thus the logic continuation of Exbrayat's work. Choosing this ground truth allow us to evaluate the role of two different hyperparameters, d and r .

In fact, we decided to only evaluate degree's role and not focus on r. We could expect that thoses parameters play equivalent role, but it would be interesant to determine if the key parameter is shape or the maximum slope of peaks. For the end of the study, we set $r = 85$. Here is a 2D single peak ground truth example with $d = 10$.

E. Model Architecture

For our purpose, which is to predict a one channel image from a three channels image, we choosen a UNet architecture, the same choice of Exbrayat.

We have not a lot of details about Exbrayat's UNet, excepted input and output size, and number of channel of the first layer, 16.

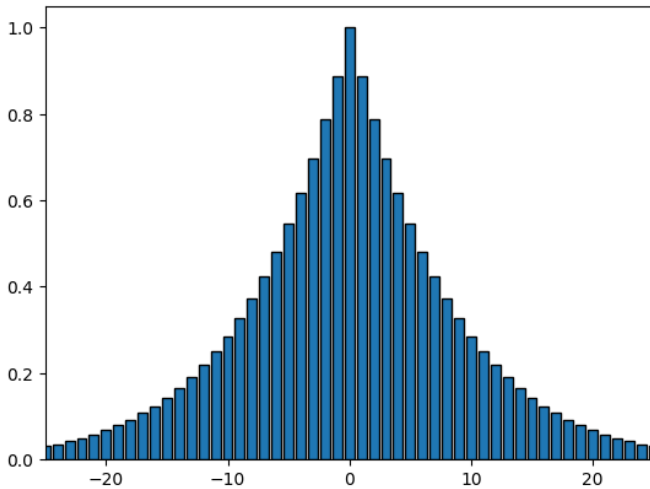


Fig. 5: Peak shape for $d = 10$, cross-section

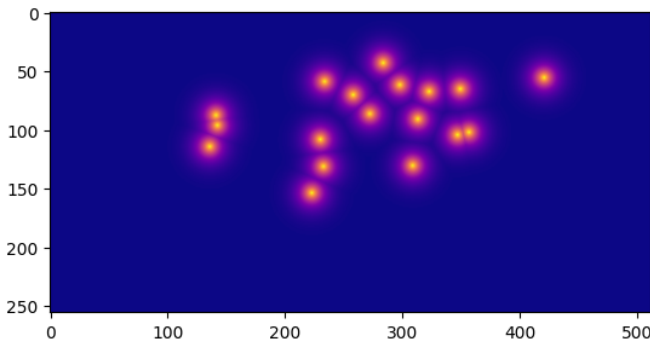


Fig. 6: ground truth example for $d=10$, values from 0 (deep blue) to 1 (warm yellow)

For our UNet, we kept this number of channels, and designed a UNet with 4 contraction phases, and 4 upsampling phases. Contractions were achieved with max poolings, and upsampling with transposed convolutional layers. Between each phases, we set 2 convolutional layers with kernel of size 3×3 . Each convolutional layer got a padding of 1, so the shape is not modified after a convolutional layer. ReLU activation functions and batch normalisations are used along the model after each convolutional layer. A `HardSigmoid` function is finally applied to go back to $[0,1]$ range for the output.

The model has been implemented with Pytorch.

F. Point recuperation and evaluation metrics

Given that the original goal was to predict 18 landmarks on a wing image, we needed a method to recover these landmarks from the predicted density map.

Exbrayat proposed to use the h -maxima transform to extract peaks, selecting the 18 most prominent points by dichotomy, i.e. finding the highest h value that yields exactly 18 h -maxima. However, in our context, this approach proved ineffective, most of the noisy peaks are located in the dark

background regions. As a result, the h -maxima method does not provide meaningful filtering.

Instead, we opted for an easier and more robust approach : extracting local maxima directly from the output map. We then select the 18 local maxima with the highest values as the predicted landmarks.

To evaluate the prediction, we must associate each predicted point with a corresponding ground truth annotation. We use the following iterative matching algorithm:

- 1) For each ground truth point i , associate all predicted points for which i is the closest ground truth. We call these the *neighbors* of i .
- 2) For each ground truth point with at least one neighbor, assign the closest neighbor as its match.
- 3) Remove all matched pairs and repeat the process with the remaining unmatched points.

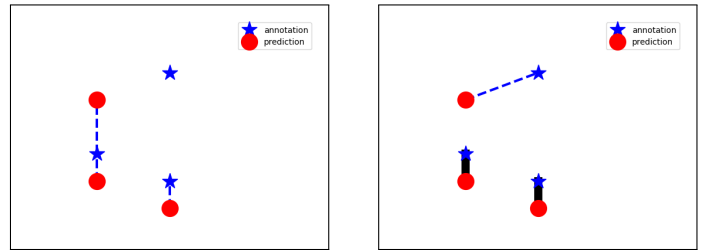


Fig. 7: algorithm illustration

To determine whether a predicted point is correct, we apply a distance threshold. Following Exbrayat, we used a threshold of 5 pixels. A predicted point is considered correct if it lies within 5 pixels of its corresponding ground truth.

However, we note that this strict threshold may underestimate the quality of predictions. In many cases, a wing may be classified as incorrectly predicted due to a single outlier point, even if the overall prediction is visually acceptable. A higher threshold could provide a more nuanced evaluation.

An alternative idea would be to incorporate the height of the local maxima into the evaluation, to reflect prediction certainty.

IV. TRAINING AND RESULTS

A. Training details

In order to train a UNet to generate a pseudo-density map from a bee wing image, we divided our dataset of 1717 images in 3 sets, the first one for train, which contain 1374 images. The second one for validation, with 171 images, and the third one for test, with 172 images. We've set the batch size to 8, and all models were trained over 150 epoch with a 20 epoch patience. We've used a `L1Loss` for training, to follow up on Exbrayat's work. ADAM was used as optimizer. We set the learning rate to 0.001 .

B. Data Augmentation

Data augmentation was used in order to help the model to generalize. In fact, data preparation already include data augmentation, thanks to the random cropping and the rotation.

However, for base 2, the different shape of wings can lead to quite variable point repartition, so to help the model to generalize on these images, we add a shift and zoom augmentation. horizontal shift range, vertical shift range and zoom range are respectively set to 0.1, 0.2 and (0.8-1.2). In his work, Mr. Exbrayat seem to change some image characteristics such as brightness or sharpness, but this doesn't appeared really necessary, so we decided not to reproduce this preprocessing.

We finally decided to add a Random resolution augmentation. This might be useful to help the model generalize on images from base 2, given that some of them got a much lower resolution than images from base 1. To do so, we perform a resize of the input image to a lower shape, then resize the image to the original shape, both using linear interpolation. Resolution lowering parameter range was set to (1-2).

C. Training results : train and validation curves

We trained several models with different value of d , respectively 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 15, 18, 22, 26 and 30. For most cases, training works well, with continuously decreasing train and validation loss. We can note that training curves appeared more noisy for low degree. An interesting phenomenon appeared for $d = 22$, with the model failing to find a way to converge and sticking to black output, and after few epoch converging to a decent solution.

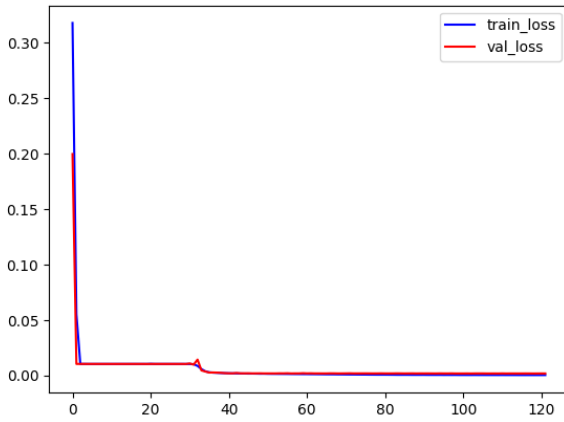


Fig. 8: training curves for degree = 22

We could expect that optimization will no more work well for higher degrees, and this was effectively the case, model with $d = 26$ learned to output a black image. However, we achieve to train a model with $d = 30$, using the already trained model $d = 8$ as initial state. With this technique, we managed to reach higher degree (see Subsection E).

D. Training results - Metrics and choice of the best parameter

We then decided to evaluate each model (excepted the model for $d = 26$, which output a black image) using metric introduced above, precision and accuracy. The first metric is the average distance for a correctly classified point from the

original annotation. The second one evaluate the part of wings which has at least one point misclassified (e.g. distance to the closest annotation greater than 5 pixels) For these metrics, we obtain the following plots.

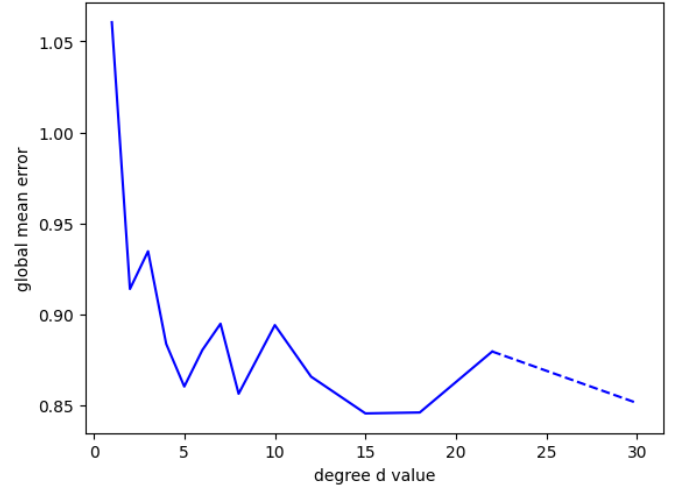


Fig. 9: global mean error (in pixel) against degree

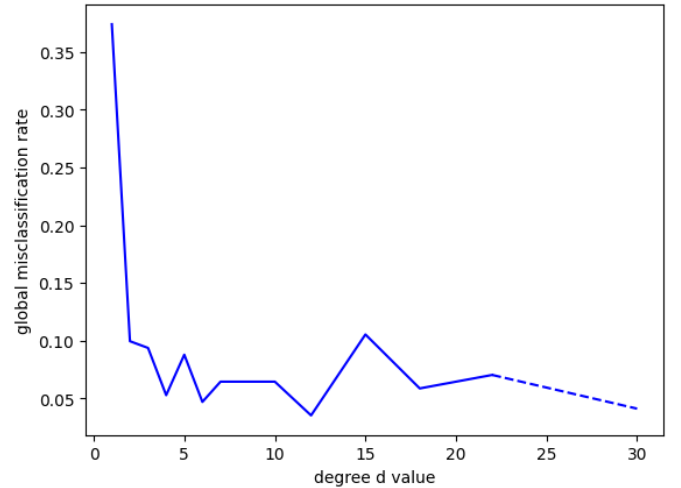


Fig. 10: misclassification rate (fraction) against degree

There are few comments that can be done about these curves. Firstly, we note that both curve decrease for degrees under 3. This is mainly because models fail to separate close landmarks for a too low degree in some cases. This effect was reported by Exbrayat in his report.

Best performing models are observed between degree 6 and degree 12. For these degrees, we have a low variance on misclassification rate, which vary from 0.03 to 0.06, and a mean error fluctuating between 0.85 and 0.90 pixels. After, it seem that models get more difficulties to converge to a high quality model, with more variability in misclassification rate. We however observe for some models a lower mean error, which mean that points correctly classified are more precisely predicted.

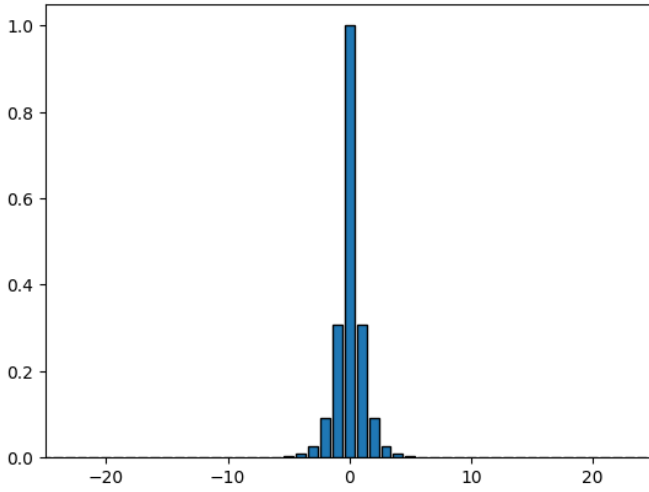


Fig. 11: Peak shape for $d = 100$, cross-section. We almost predict a single point.

As mentionned before, point for degree 30 is quite special, because this model is obtained from degree 8 model. We note for this model a decrease in both metric compared to the degree 8 model. We investigate the effects of this technique on section E. The improvement might be caused by the narrowing of peaks, which allow the model to really output the points it want to predict, and less be subject to fluctuations of output peaks shape. In other words, this give more importance to the point we want the model to predict, and less to the peak shape. This is thus more adapted to our point recovery process, which only uses a single point to retrieve predicted annotation. At this point model with degree 30 is the best performing one on validation set, with 96 % accuracy (164 correctly classified wings for 7 misclassifications) and 0.85 pixels mean error with threshold equal to 5.

E. Training results - Finetuning to increase degree

To further investigate on the effects of increasing degree through finetuning. We decided to do the following protocol. We start from the model trained with degree 1 ground truth from the precedent experiment. We then increased gradually the degree, training each new model from the preceding one over 20 epoch. In fact, only 1 or 2 epoch are necessary to models to adapt peak shape. For this experiment, we set degree's values respectively to 2, 3, 5, 8, 11, 14, 18, 22, 27, 33, 39, 46, 60 and 100.

We evaluated the results with the same protocol as mentionned before. We then obtained the curves below.

Some phenomena must be mentionned. First, we have noticed that noise in the black part of the output tend to dissapear when degree increase (and interations of the process). This has caused some issues because for few cases, some models (degree = 11, 14, 18, 22, 46, 60) outputs got only 17 local maxima (when close landmarks were merged). Second, even if we note an overall improvement in performances when degree increase. This is not the case when we take point separately.

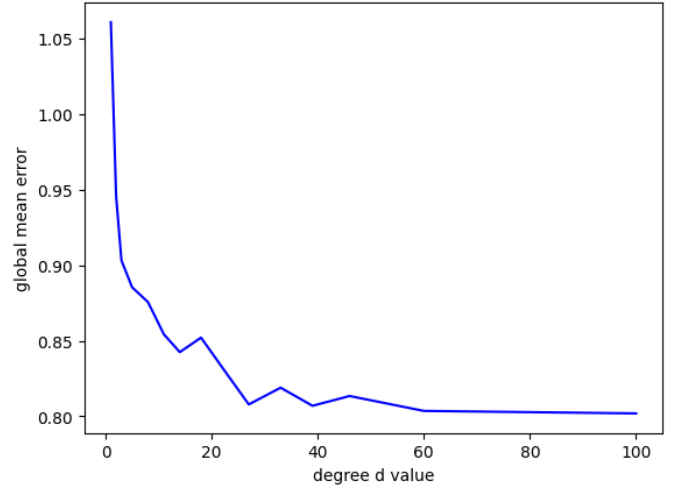


Fig. 12: global mean error (in pixel) against degree for finetuned models

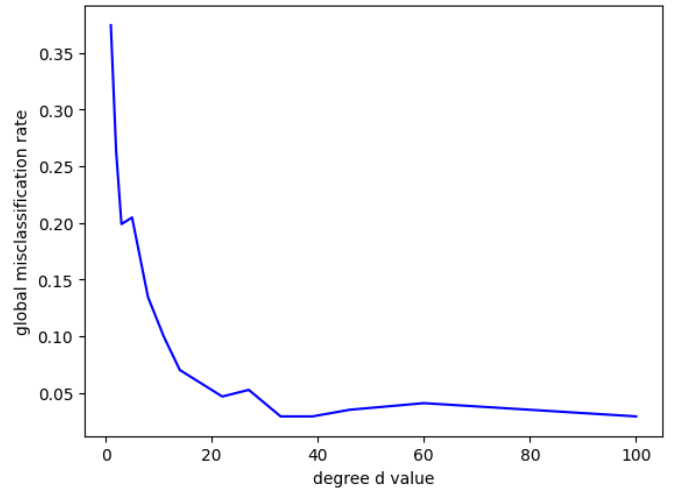


Fig. 13: misclassification rate (fraction) against degree for finetuned models

For instance model with degree 100 is the best performing on point 17 but is less accurate than the other ones on points 16, even if it inherit from their structure.

We however reached very good performances for some models, the best being the one for degree 100. A logic continuation of the work would be to determine how to mix up models to get better performances. At this point, the best model if model for degree 100, with 97 % accuracy (167 correctly classified wings for 4 misclassifications) and 0.80 pixels mean error with threshold equal to 5. We can observe in both training sessions that, at some point, the mean error start to flatten. This may indicate that an optimum has been reached, potentially due to inherent variations in the annotation process.

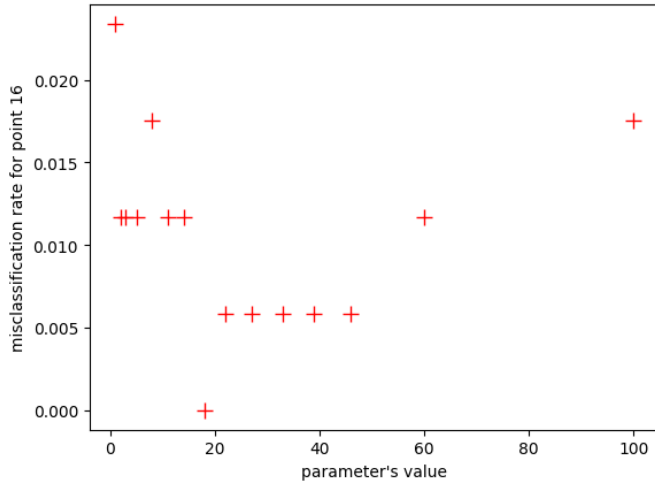


Fig. 14: misclassification rate against degree for point 16, finetuned models

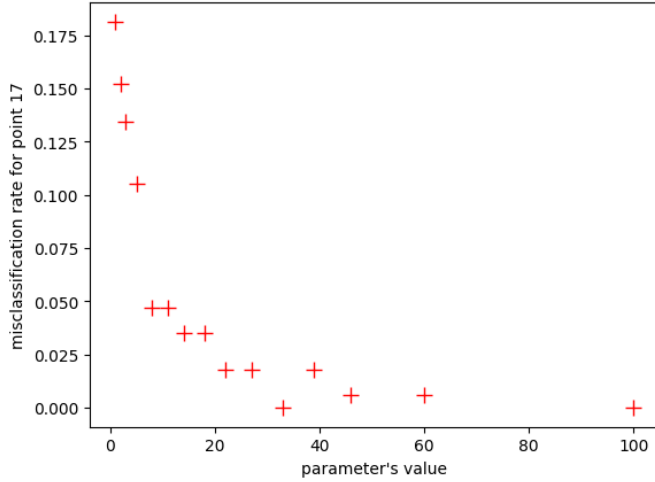


Fig. 15: misclassification rate against degree for point 17, finetuned models

F. Testing on test set

We can now test our model with its optimal parameters. For most points, our model perform really well, with often predicted points within a distance of 1 pixel to the ground truth. However, some points remain problematic, such as point 0. For this point, the prediction is the less precise among all the points. This can be clearly explained by the fact that this point is in a zone where veins have often not well defined contours. We can also note that conversely to Exbrayat's final results, close landmarks no more really represent a problem, with only very few misclassifications of these points (points 11 and 17).

On test set, our best model obtained a 97 % accuracy score, and 0.83 pixels mean error. Note that it also had all its predictions within 8 pixels to the original ones, which would have made 100 % accuracy with a threshold of 8.

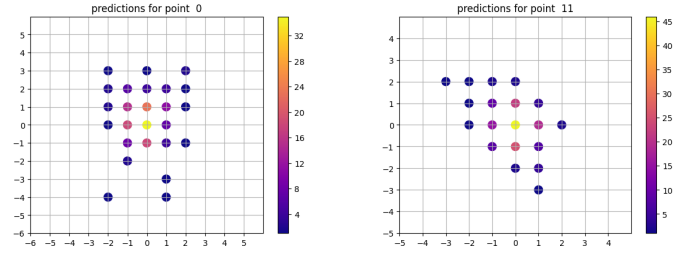


Fig. 16: predictions distribution on test set for points 0 and 11
Distance Range and Density (log scale) for distance from 0 to 15 pixels

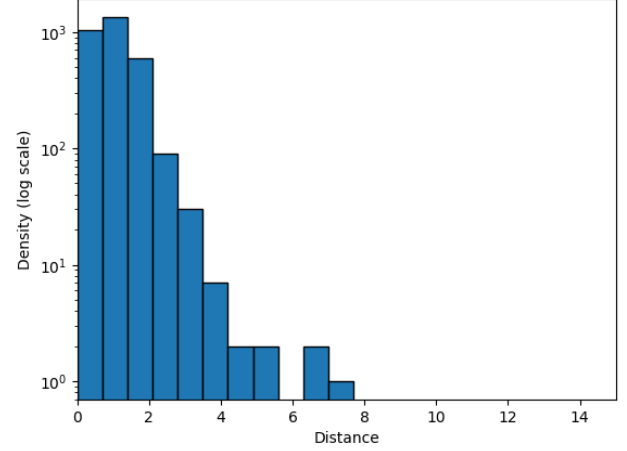


Fig. 17: density against distance to the original annotation on test set for degree 100 model, note that there are no predictions farther than 8 pixels to the original ones.

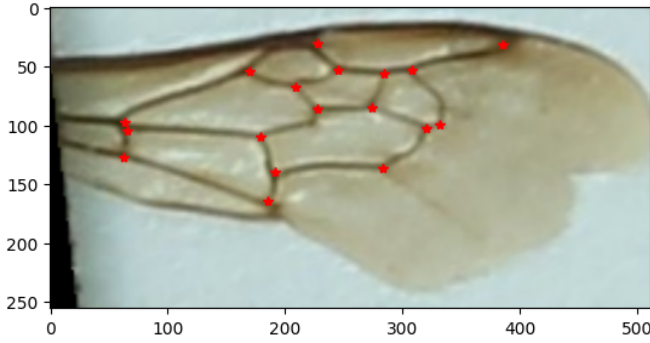
G. Testing on base 2

For testing on base 2, we have extracted and prepared 258 images from base 2. Given that this base is unannotated, we do not have a way to directly evaluate precision for a prediction. We however could do a visual evaluation of results, which is enough to have an estimation of accuracy. Our model correctly predicted 18 points for each prepared image (e.g. all predicted points were approximately located where we would expect to find them) We however noticed few phenomena that would help to evaluate prediction's quality.

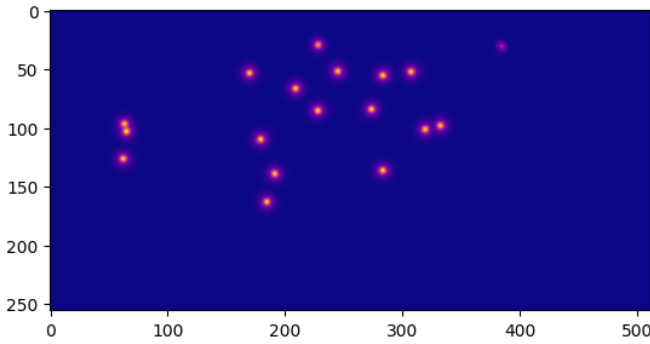
First, for some wings, the model failed to deliver a consistent output when the input is the segmented wing on a black background. After scrutiny, this was caused by the bad quality of segmentation, the segmentation was too tight and didn't captured some useful parts in the border of some wings. We thus decided to only evaluate performances on cropped wing (and so we kept original background).

Second, On some wings, some points are weakly predicted, which mean that some predicted peaks are not well defined (noisy shape, maximum far from 1). This led us to compute a reliability score based on this observation. To compute this score, we first generated a fake ground truth image based on predicted points. We then computed the mean squared distance between the output of the model and this fake ground truth. We obtained a score, that we calculated for each prepared

image from base 2. We could then set a threshold to exclude prediction with a too high score. Here is an example of image with high score, with its predicted ground truth.



(a) prediction made on base 2 prepared wing



(b) output of the model, we can note that the peak corresponding to point 0 is quite weak

Fig. 18: prediction and output for a high score wing

This could be a good solution to evaluate the quality of a prediction for unsupervised cases. For future works, this would be interesting to determine if this phenomenon also appears on base 1 when predictions are unprecise.

V. CONCLUSION AND PERSPECTIVES

A. Conclusion

The goal of this paper was to further investigate on point prediction on bee wings, in order to improve performances obtained by V. Exbrayat in his work. For this purpose, we have decided to follow up on Exbrayat's work, adapting his technique with different ground truths. We show that for a single channel output, higher degree allowed better accuracy and precision. We show too that models could predict ground truth with very high degree peaks, using a technique of fine-tuning. We have noticed that this would however cause worst performances for some landmarks non-problematic for lower degrees. To predict the 18 landmarks, Exbrayat achieved 77 % accuracy and 1.21 pixels mean error. With our improvements, we achieved to reach 97 % accuracy and 0.83 pixels mean error. Note that we have also changed the data preparation to reach these performances, with a limitation in wing angle. On the other side, Exbrayat excluded more data from the

annotated dataset, which could partly explain why he got worst performances. We have tested our best model on some images from unannotated base 2, after designing a special data preparation using SAM. Results were visually satisfying, even if some outputs were not as expected, with some weak peaks, which might be the expression of an unprecise prediction.

B. Perspectives

Several additional testings were done to evaluate the viability of some directions. We could have investigated on the possibility of several layers output, or single layer output mixing several degrees, to adapt to each point specificity (prediction variability). We could imagine that low degree peaks are more adapted to predict points with high variability (point 0), and higher degree peaks more adapted for low variability points (points 3, 12).

Single layer output with different degree peaks might be tricky, and might cause some issues on prediction's quality when degree range is too high. Several layers output is also challenging because of memory issues and material limitations.

A good trade off would be to mix single layer output models to get the best from each of them. We will then have a model with multiple UNet in parallel.

Experiments show that UNet was able to predict a single point from the wing. This could be a good direction to investigate to improve the reliability of outputs.

We have also barely investigated on the possibility to segment cells in bee wings using SAM, for images from base 2. We had to abandon this approach because it was not working on annotated base (grid in the background) and we had not a way to determine if this would give good point predictions on base 2 wings. It would however be interesting to improve the way the algorithm is implemented, and to add the point prediction from the cells predictions, to have another way to predict points and compare to the first one on unannotated dataset.

Finally, we have reshaped some images so that they could fit in the network, but we could imagine to adapt the network so that it could deal with higher resolution wings and thus have better performances. For this purpose, we could imagine to upscale images instead of lowering their quality.

REFERENCES

- [1] V. Exbrayat 'Identification of bumblebee species based on their wings', 2024, Rapport de projet, Paris, Mines Paris-PSL, Centre de Morphologie Mathématique, Unavailable
- [2] Toulzac, M., M. Methlouthi A. Perrard (2022). 'Soda maker for field anesthesia as a step towards a non-lethal identification of wild bees and other flower visitors. Osmia, 10: 25–34. -OSMIA- Revue d'Hyménoptérologie - Journal of Hyme', OSMIA. Accessed: May 07, 2024. (Online). Available: <https://www.osmia-journal-hymenoptera.com/osmia-10-3.html>
- [3] M. BAYLAC, C. VILLEMANT, and G. SIMBOLOTTI, 'Combining geometric morphometrics with pattern recognition for the investigation of species complexes', Biological Journal of the Linnean Society, vol.80, no. 1, pp. 89–98, Sep. 2003, doi: 10.1046/j.1095-8312.2003.00221.x.
- [4] K. Buschbacher, D. Ahrens, M. Espeland, and V. Steinhage, 'Image based species identification of wild bees using convolutional neural networks', Ecological Informatics, vol. 55, p. 101017, Jan. 2020, doi: 10.1016/j.ecoinf.2019.101017.

- [5] C. Pinho, A. Kaliontzopoulou, C. A. Ferreira, and J. Gama, 'Identification of morphologically cryptic species with computer vision models: wall lizards (Squamata: Lacertidae: Podarcis) as a case study', *Zoological Journal of the Linnean Society*, vol. 198, no. 1, pp. 184–201, May 2023, doi: 10.1093/zoolinnean/zlac087.
- [6] A. Perrard, 'Systématique et morphométrie géométrique: l'évolution de la nervation alaire au sein du genre *Vespa* (Hyménoptères, Vespidae)', *These de doctorat*, Paris, Muséum national d'histoire naturelle, 2012. Accessed: May 14, 2024. (Online). Available: <https://theses.fr/2012MNHN0026>
- [7] O. Ronneberger, P. Fischer, and T. Brox, 'U-Net: Convolutional Networks for Biomedical Image Segmentation', presented at the LNCS, Oct. 2015, pp. 234–241. doi: 10.1007/978-3-319-24574-42_8.
- [8] Lempitsky, V., Zisserman, A. (2010). Learning to count objects in images. *Advances in neural information processing systems*, 23
- [9] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollar, Ross Girshick, 'Segment Anything', *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 4015–4026